

Mecklenburg County Sample



Traffic Impact Study

Prepared September 2, 2026 · 35.1954, -80.7643

Simple Impact Studies — sample draft

Traffic Impact Study

Generated 2026-09-02T13:32:43.834Z

EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) evaluates the anticipated traffic impacts of the proposed Mecklenburg County Sample within Charlotte MSA, North Carolina (City of Charlotte / Mecklenburg County). The analysis follows the structure of the NCDOT Policy on Street and Driveway Access to North Carolina Highways (July 2003) Chapter 5 and the NCDOT Congestion Management Capacity Analysis Guidelines (Standards, March 2022; Best Practices, November 20, 2024). 15 study intersections within a 1.00-mile radius are analyzed under Existing, No-Build, and Build conditions.

15	0	11	6,800
INTERSECTIONS	LOS DROPS	AT LOS E/F	DAILY TRIPS (RAW)

1.0 INTRODUCTION

Project name	Mecklenburg County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	35.1954°, -80.7643°
Build-out (horizon) year	2027
Controlling jurisdiction	City of Charlotte / Mecklenburg County
MPO	CRTPO (Charlotte Regional Transportation Planning Organization) — 2026-2035 TIP
Programmed projects source	NCDOT 2026-2035 STIP (adopted July 2025), as amended

1.1 TIA Applicability

Charlotte UDO Article 32, Section 32.1 (effective 6/1/2023) requires a Comprehensive Transportation Review (CTR) — Multimodal Assessment + TDM Assessment + Traffic Impact Study — when Charlotte Streets Manual Chapter 3, Table 3.1 thresholds are met (reported secondary-source values: >1,500 vpd or >150 peak-hour trips in lower-intensity districts; >2,000 vpd or >200 peak-hour trips in medium/high-intensity districts — confirm against the current Table 3.1 at scoping). The CTR applies at permitting even for by-right development.

The proposed development's estimated raw trip generation of 6,800 vehicles per day meets or exceeds the 3,000 vpd threshold at which NCDOT may require a TIA (no reductions are applied in the threshold test).

1.2 Scoping

CDOT publishes a Rezoning Transportation Analysis memo per rezoning petition; the TIS must be prepared by a North Carolina licensed Professional Engineer. NCDOT coordination applies on state-system streets. The background growth factor and the list of approved background developments must be agreed in the Approved Scoping Document prior to TIA submittal; a placeholder for that document is carried in the Appendix.

2.0 METHODOLOGY

NCDOT requires HCM-based analysis at submittal; the Department currently utilizes Synchro 11 (with mandatory SimTraffic simulation, minimum ten runs), HCS for freeway facilities, and Sidra for roundabouts. This screening applies the openly-published Webster/Akçelik signalized model pending those tool runs; Synchro 11-compatible models and SimTraffic outputs are to be provided at formal submittal.

Peak periods	AM and PM weekday peak hours (minimum); counts Tue–Thu, school in session, less than 12 months old
PHF (future conditions)	0.90 per Standards
Heavy vehicles	Average of duals and TTSTs, 2% minimum
RTOR (future analysis)	Not permitted per Standards
Turn-lane storage	Greater of Synchro 95th percentile and SimTraffic maximum queue, rounded up to 25 ft; 100 ft minimum
Roundabout threshold	v/c 0.85
Study-area rule	Intersections where site trips add 10% or more to background volumes on any approach/movement

2.1 Trip Generation Basis

NCDOT's Best Practices call for the ITE Trip Generation Manual with the Department's Rate-vs-Equation spreadsheet, internal capture per the NCHRP 684 spreadsheet method (vehicle occupancy 1.1, 4,000-ft maximum walking distance), and pass-by limited to retail uses with multi-use pass-by capped at 10% of the adjacent street volume. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal; applied internal-capture and pass-by credits must be reconciled against the NCDOT limits at scoping.

Daily rate applied	80.00 trips per ksf × 85 ksf
AM peak rate applied	3.20 trips per ksf × 85 ksf
PM peak rate applied	8.00 trips per ksf × 85 ksf
Rate source	SANDAG 2002 “(Not So) Brief Guide of Vehicular Traffic Generation Rates for the San Diego Region” — Community Shopping Center (daily 80/ksf; PM pass-by 30%)

3.0 ANALYSIS CONDITIONS AND MEASURES OF EFFECTIVENESS

Scenarios follow the NCDOT Standards set: Existing Base Year; No-Build Design Year (build-out); and Design Year Build. Where a scoped project is phased, intermediate build-out years are analyzed with earlier-phase trips carried as site trips (never background). Where an impacted STIP project is in planning, design, or construction (or within 5 years post-construction) and site traffic was not in the STIP forecast, an additional STIP design-year analysis is coordinated per the Best Practices TIA/STIP flow chart.

3.1 Intersection MOE Summary (NCDOT format)

Approaches are ordered EB, WB, NB, SB and movements left-through-right per the Best Practices reporting format. Signalized intersections report control delay and LOS overall and per lane group (any v/c > 1.0 is reported as LOS F regardless of delay); unsignalized intersections report no overall LOS — results are per lane group with the conflicting movement. SimTraffic maximum-queue columns are populated at submittal.

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
1	North Sharon Amity Road & Holbrook Drive	B / 14.6s	B / 14.6s	B / 15.6s
2	North Sharon Amity Road & Pierson Drive	F / 234.0s	F / 243.6s	F / 288.1s
3	North Sharon Amity Road & Charleston Drive	B / 16.3s	B / 16.3s	B / 16.7s
4	Albemarle Road & North Sharon Amity Road	F / 300.0s	F / 300.0s	F / 300.0s
5	Albemarle Road & Pierson Drive	B / 14.1s	B / 14.1s	B / 14.2s

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
6	North Sharon Amity Road & Monroe Road	F / 212.0s	F / 221.2s	F / 233.6s
7	Ashmore Dr & Monroe Rd	F / 116.1s	F / 123.3s	F / 137.6s
8	Idlewild Rd & Monroe Rd_rama Rd	F / 96.4s	F / 103.0s	F / 110.0s
9	Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	F / 300.0s	F / 300.0s	F / 300.0s
10	Idlewild Road & Connection Point Boulevard	E / 70.1s	E / 75.2s	E / 76.9s
11	Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	F / 300.0s	F / 300.0s	F / 300.0s
12	Mcclintock Middle School	F / 116.1s	F / 123.3s	F / 124.3s
13	Flasher Sheffield Dr & Woodland Dr	F / 300.0s	F / 300.0s	F / 300.0s
14	Oakhurst Elementary School	B / 14.4s	B / 14.4s	B / 14.5s
15	Albemarle Rd & Reddman Rd	F / 300.0s	F / 300.0s	F / 300.0s

3.2 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^{1.1} \cdot d_{site})$); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector		Share of project trips			
Northeast (NNE + ENE)		44%			
Southeast (ESE + SSE)		15%			
Southwest (SSW + WSW)		14%			
Northwest (WNW + NNW)		27%			

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Flasher Sheffield Dr & Woodland Dr	WNW	0.93	11,250	0.3	25.69%
Albemarle Rd & Reddman Rd	ENE	0.98	4,365	0.1	10.34%
Albemarle Road & North Sharon Amity Road	NNE	0.47	3,510	0.1	9.82%
Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	NNE	0.83	3,510	0.1	8.74%
North Sharon Amity Road & Pierson Drive	NNE	0.14	2,610	0.1	8.26%
Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	NNE	0.86	2,880	0.1	7.11%
North Sharon Amity Road & Monroe Road	WSW	0.61	2,520	0.1	6.73%
Ashmore Dr & Monroe Rd	SSW	0.64	2,115	0.1	5.60%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Mcclintock Middle School	SSE	0.90	2,115	0.1	5.15%
Idlewild Rd & Monroe Rd_rama Rd	SSE	0.80	2,025	0.1	5.10%
Idlewild Road & Connection Point Boulevard	SSE	0.85	1,890	0.0	4.67%
North Sharon Amity Road & Charleston Drive	WSW	0.19	477	0.0	1.48%
North Sharon Amity Road & Holbrook Drive	WNW	0.02	198	0.0	0.65%
Oakhurst Elementary School	WSW	0.97	162	0.0	0.39%
Albemarle Road & Pierson Drive	WNW	0.58	100	0.0	0.27%

15 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

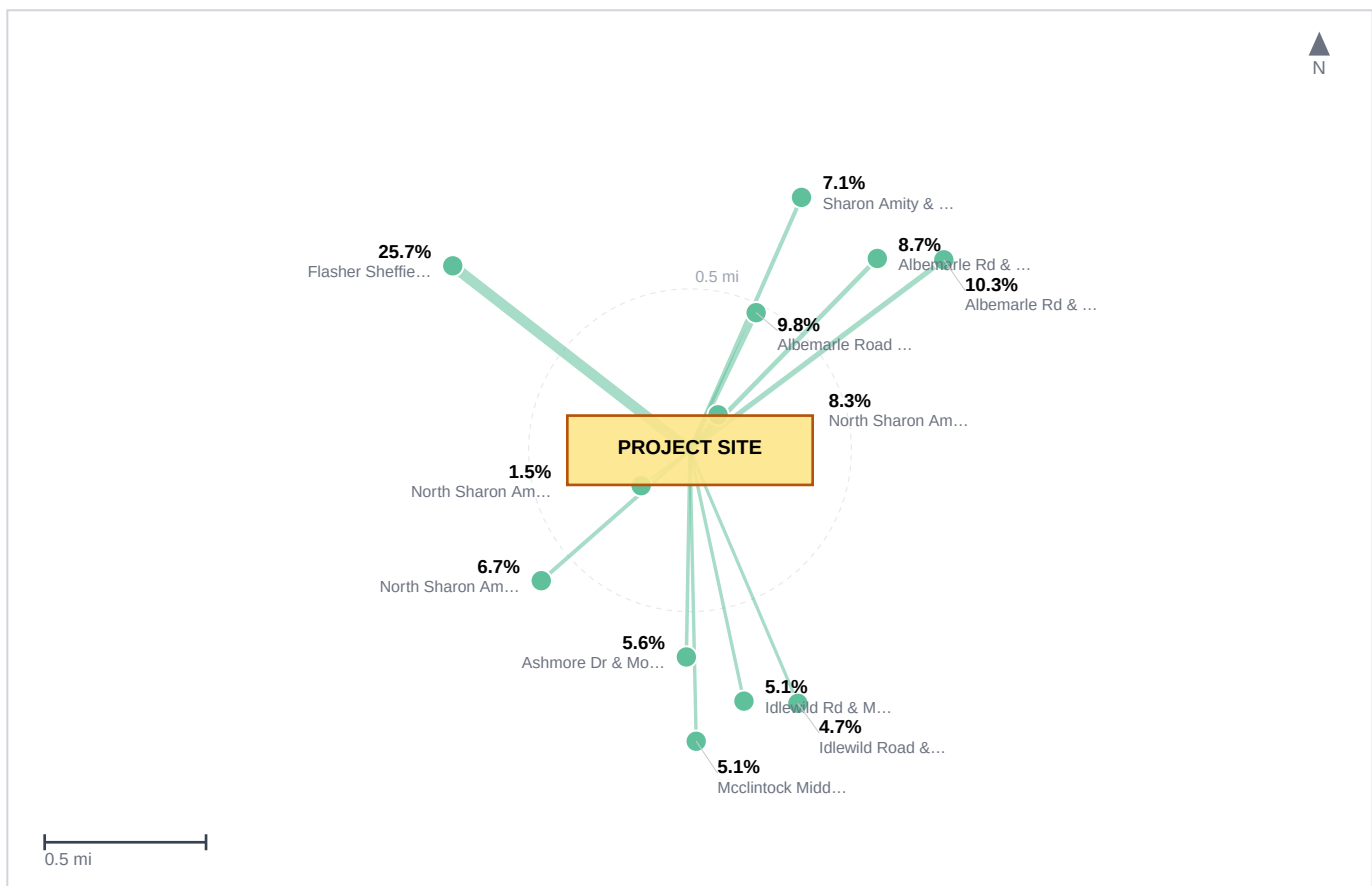
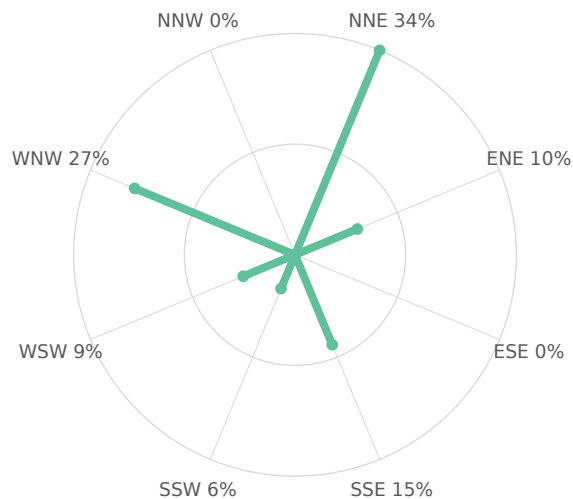


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

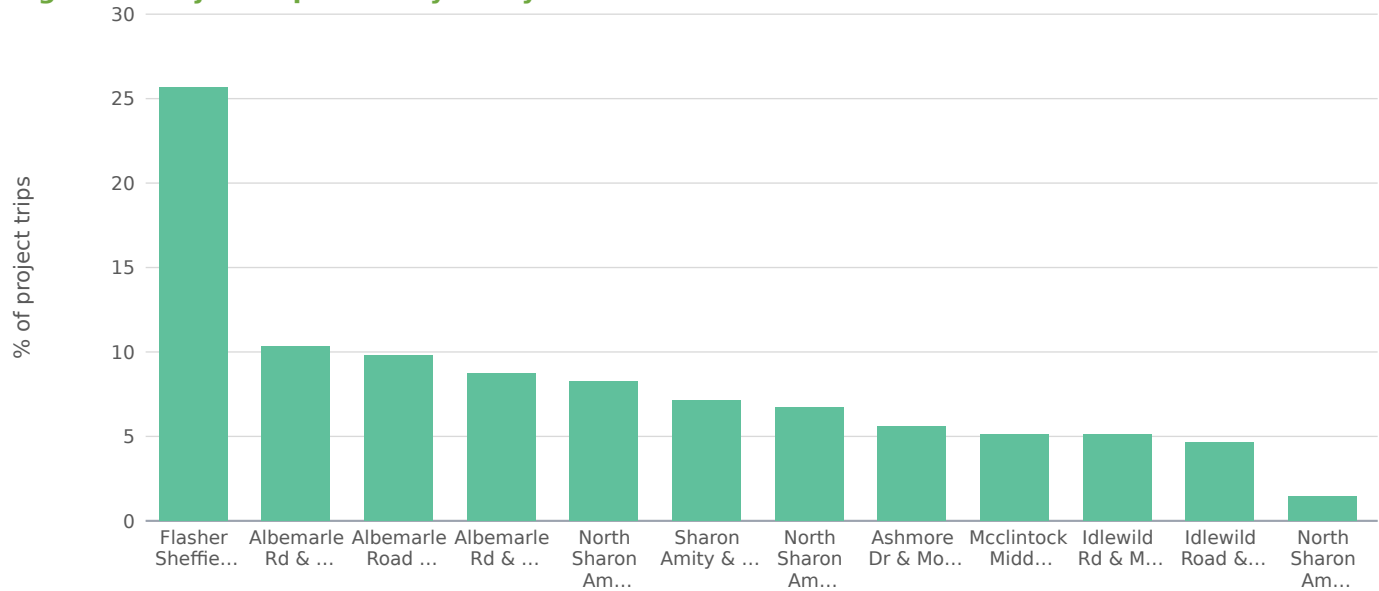
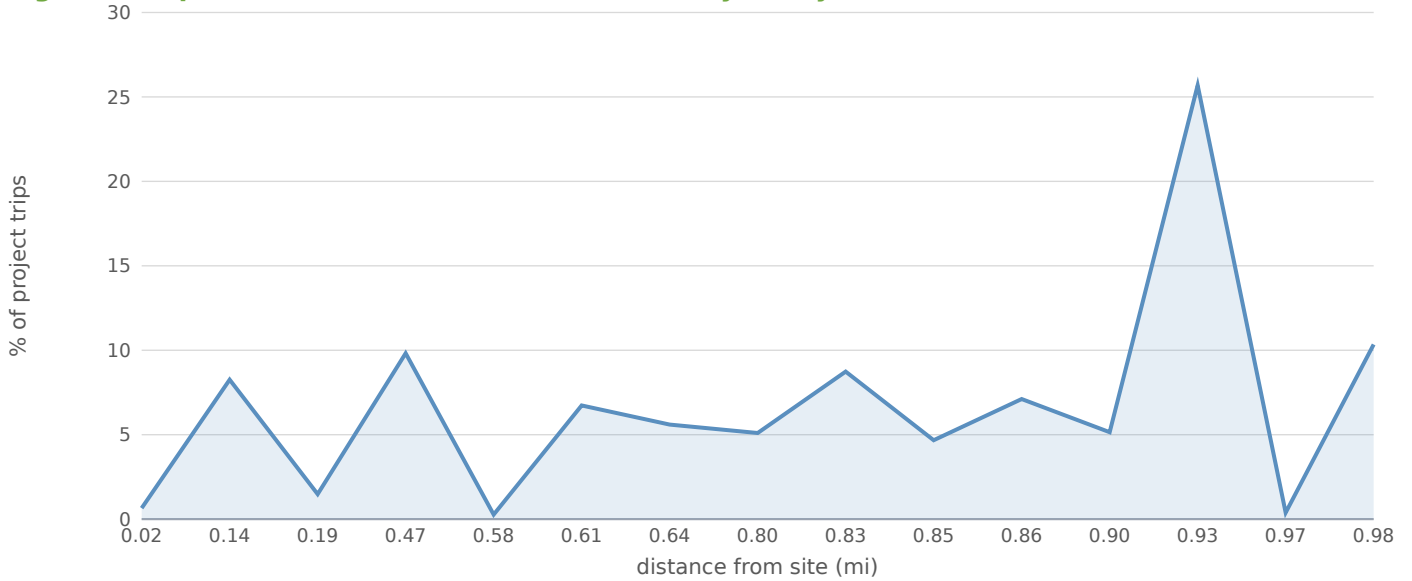


Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.3 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
North Sharon Amity Road & Holbrook Drive	WNW	0.02	119	177	20.6%
North Sharon Amity Road & Pierson Drive	NNE	0.14	107	181	21.1%
North Sharon Amity Road & Charleston Drive	WSW	0.19	22	55	6.4%
Albemarle Road & North Sharon Amity Road	NNE	0.47	59	89	10.4%
Albemarle Road & Pierson Drive	WNW	0.58	7	17	2.0%
North Sharon Amity Road & Monroe Road	WSW	0.61	34	51	5.9%
Ashmore Dr & Monroe Rd	SSW	0.64	33	62	7.2%
Idlewild Rd & Monroe Rd_rama Rd	SSE	0.80	13	32	3.7%
Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	NNE	0.83	3	6	0.7%
Idlewild Road & Connection Point Boulevard	SSE	0.85	6	9	1.0%
Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	NNE	0.86	75	141	16.4%
Mcclintock Middle School	SSE	0.90	2	4	0.5%
Flasher Sheffield Dr & Woodland Dr	WNW	0.93	2	4	0.5%
Oakhurst Elementary School	WSW	0.97	7	19	2.2%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Albemarle Rd & Reddman Rd	ENE	0.98	6	11	1.3%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

4.0 MITIGATION CRITERIA CHECK (POLICY CH. 5.J)

NCDOT requires improvements when, comparing base-network and project conditions: total average delay increases by 25% or more while remaining at the same LOS; the LOS degrades by one level; or the intersection operates at LOS F. Turn-lane mitigation applies where the 95th-percentile queue exceeds existing storage. Signal-timing changes alone are not considered mitigation. The District Engineer makes the final mitigation determination.

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
1	North Sharon Amity Road & Holbrook Drive	+7%	no	no	none indicated
2	North Sharon Amity Road & Pierson Drive	+18%	no	YES	MITIGATION INDICATED
3	North Sharon Amity Road & Charleston Drive	+2%	no	no	none indicated
4	Albemarle Road & North Sharon Amity Road	+0%	no	YES	MITIGATION INDICATED
5	Albemarle Road & Pierson Drive	+1%	no	no	none indicated
6	North Sharon Amity Road & Monroe Road	+6%	no	YES	MITIGATION INDICATED
7	Ashmore Dr & Monroe Rd	+12%	no	YES	MITIGATION INDICATED
8	Idlewild Rd & Monroe Rd_rama Rd	+7%	no	YES	MITIGATION INDICATED
9	Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	+0%	no	YES	MITIGATION INDICATED
10	Idlewild Road & Connection Point Boulevard	+2%	no	no	none indicated
11	Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	+0%	no	YES	MITIGATION INDICATED
12	Mcclintock Middle School	+1%	no	YES	MITIGATION INDICATED
13	Flasher Sheffield Dr & Woodland Dr	+0%	no	YES	MITIGATION INDICATED
14	Oakhurst Elementary School	+1%	no	no	none indicated
15	Albemarle Rd & Reddman Rd	+0%	no	YES	MITIGATION INDICATED

5.0 CONCLUSIONS AND SUBMITTAL NOTES

0 intersections project a LOS drop and 11 operate at LOS E or F under Build conditions; mitigation per Policy Ch. 5.J should be developed with the District Engineer.

Data conventions: NCDOT forecasts supply AADT, % duals and TTSTs, directional split (D), and design-hour factor (K); forecast conversion to peak-hour volumes uses the Intersection Analysis Utility (IAU) in the NCDOT Traffic Engineering Suite. AADT reference: NCDOT ArcGIS AADT mapping application. Crash analysis: a site-radius crash assessment from NCDOT's TEAAS (Traffic Engineering Accident Analysis System) is prepared at submittal — not generated by this screening tool.

This TIA is to be prepared under the direct charge of and sealed by a licensed North Carolina Professional Engineer with expertise in traffic engineering (Policy Ch. 5.A; 21 NCAC 56 .1103 — seal with signature over or adjacent to the seal and date of signing; firm name, address, and NC firm license number on the title block). TIAs marked confidential remain so until a driveway permit is requested or the project is publicly announced.

SCREENING DISCLOSURE: this document is a screening-level draft. It does not include the Approved Scoping Document, SimTraffic outputs, Synchro models, TEAAS crash analysis, or signed/sealed certification — all required for a formal NCDOT TIA submittal.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
North Sharon Amity Road & Pierson Drive	0.14	0.3	181	21.1%
North Sharon Amity Road & Holbrook Drive	0.02	0.1	177	20.6%
Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	0.86	2.1	141	16.4%
Albemarle Road & North Sharon Amity Road	0.47	1.1	89	10.4%
Ashmore Dr & Monroe Rd	0.64	1.5	62	7.2%
North Sharon Amity Road & Charleston Drive	0.19	0.5	55	6.4%
North Sharon Amity Road & Monroe Road	0.61	1.5	51	5.9%
Idlewild Rd & Monroe Rd_rama Rd	0.80	1.9	32	3.7%
Oakhurst Elementary School	0.97	2.3	19	2.2%
Albemarle Road & Pierson Drive	0.58	1.4	17	2.0%
Albemarle Rd & Reddman Rd	0.98	2.4	11	1.3%
Idlewild Road & Connection Point Boulevard	0.85	2.0	9	1.0%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 84% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 16% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 86.7% of study signals lie on a modeled route; highest loaded-link v/c : 0.59. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Minor arterial	3.76	380	0.35
Trunk highway	1.60	142	0.29
Collector	0.18	132	0.59

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 16 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 492 network junctions, max imbalance 0.00 veh. 12 study intersection(s) carry path-derived movements; 3 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

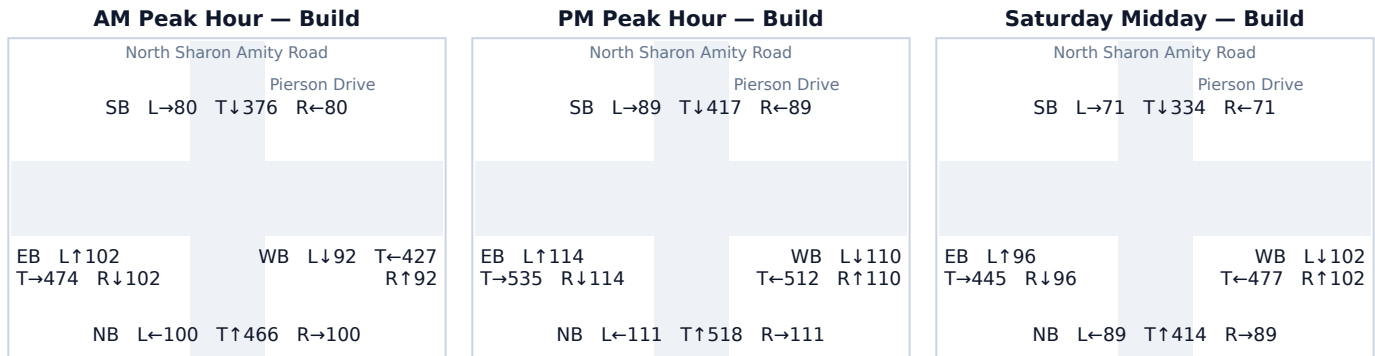
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 North Sharon Amity Road & Pierson Drive

Signal ID	charlotte-2181076118
Location	NPA 16 · 0.14 mi from site
Added PM peak trips	181
Existing / No-Build (PM)	LOS F · 243.6 s/veh · v/c 1.47
Build (PM)	LOS F · 288.1 s/veh · v/c 1.57
Δ control delay	+44.5 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	740	0	740	0.91 → 0.91	39.6 → 39.6	D → D	713
SB	595	0	595	0.74 → 0.74	26.2 → 26.2	C → C	505
EB	726	38	763	0.90 → 0.94	37.4 → 44.0	D → D	752
WB	588	143	732	0.73 → 0.90	25.8 → 38.3	C → D	699

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	143	79%
EB Left	38	21%

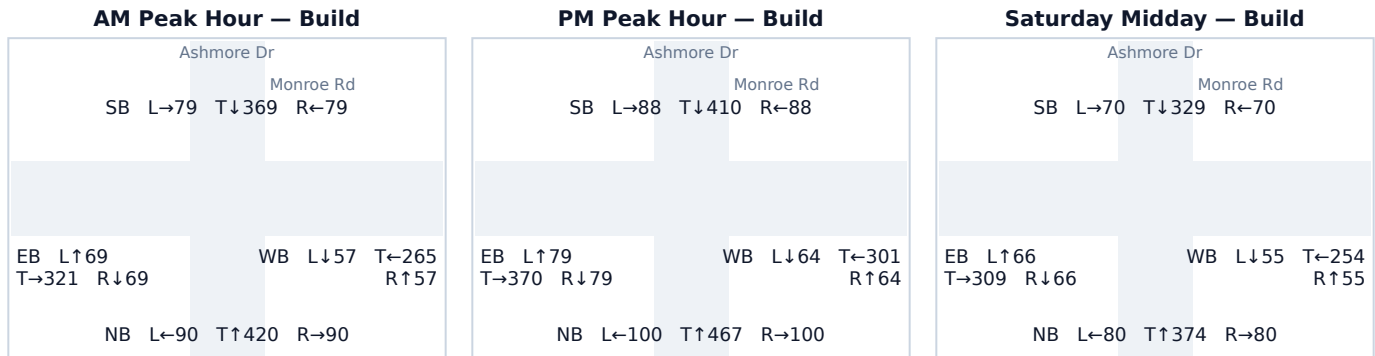
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Ashmore Dr & Monroe Rd

Signal ID	charlotte-cdot-1896
Location	NPA 52 · 0.64 mi from site
Added PM peak trips	62
Existing / No-Build (PM)	LOS F · 123.3 s/veh · v/c 1.19
Build (PM)	LOS F · 137.6 s/veh · v/c 1.23
Δ control delay	+14.3 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	667	0	667	0.82 → 0.82	30.9 → 30.9	C → C	601
SB	586	0	586	0.72 → 0.72	25.7 → 25.7	C → C	493
EB	495	32	528	0.61 → 0.65	22.2 → 23.3	C → C	423
WB	399	30	429	0.49 → 0.53	19.6 → 20.3	B → C	319

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	32	52%
WB Through	30	48%

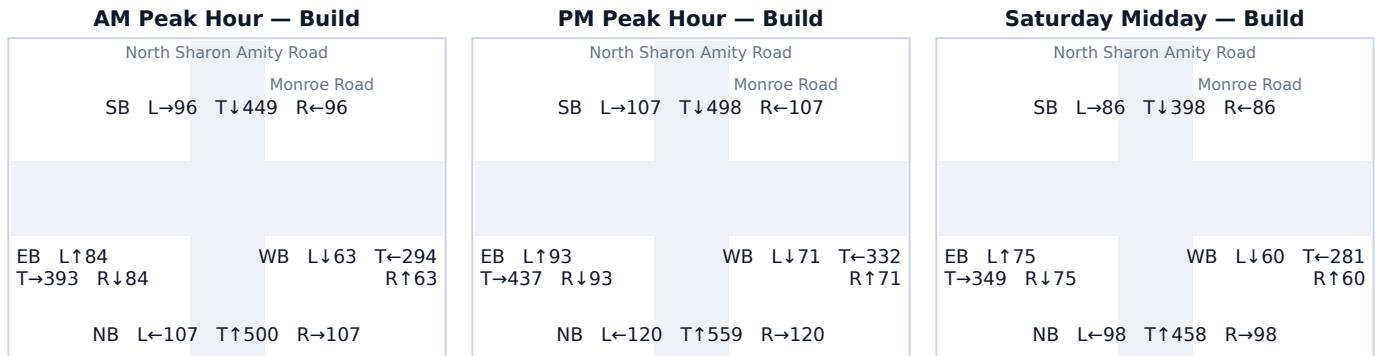
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 North Sharon Amity Road & Monroe Road

Signal ID	charlotte-5527699422
Location	NPA 52 · 0.61 mi from site
Added PM peak trips	51
Existing / No-Build (PM)	LOS F · 221.2 s/veh · v/c 1.42
Build (PM)	LOS F · 233.6 s/veh · v/c 1.45
Δ control delay	+12.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	778	21	799	0.96 → 0.99	47.4 → 53.0	D → D	815
SB	712	0	712	0.88 → 0.88	35.5 → 35.5	D → D	668
EB	623	0	623	0.77 → 0.77	27.8 → 27.8	C → C	541
WB	445	30	474	0.55 → 0.59	20.7 → 21.6	C → C	365

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	30	59%
NB Through	21	41%

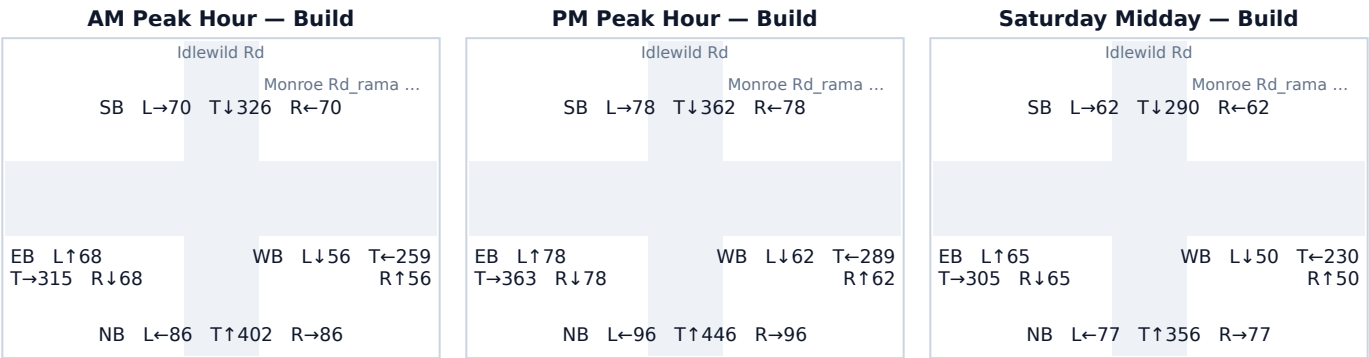
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Idlewild Rd & Monroe Rd_rama Rd

Signal ID	charlotte-7945928189
Location	NPA 52 · 0.80 mi from site
Added PM peak trips	32
Existing / No-Build (PM)	LOS F · 103.0 s/veh · v/c 1.14
Build (PM)	LOS F · 110.0 s/veh · v/c 1.16
Δ control delay	+7.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	638	0	638	0.79 → 0.79	28.7 → 28.7	C → C	561
SB	518	0	518	0.64 → 0.64	23.0 → 23.0	C → C	412
EB	487	32	519	0.60 → 0.64	21.9 → 23.0	C → C	414
WB	413	0	413	0.51 → 0.51	19.9 → 19.9	B → B	304

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	32	100%

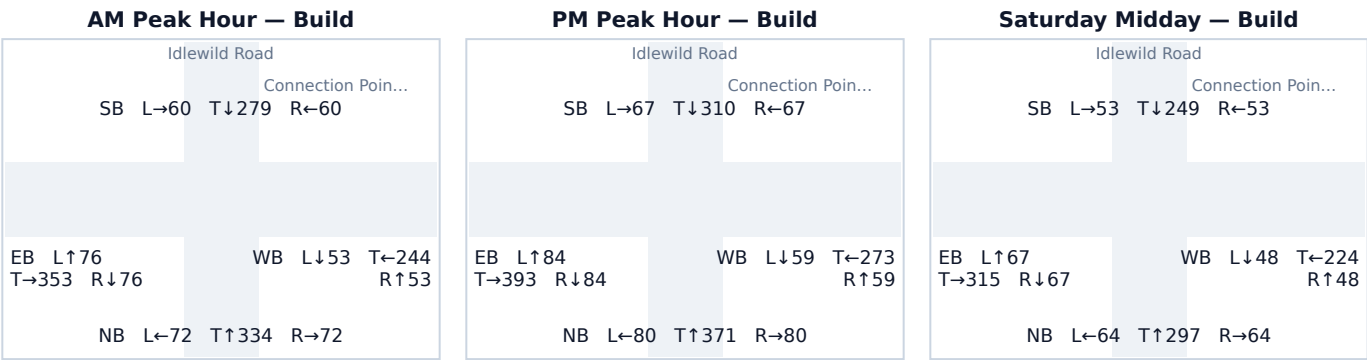
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Idlewild Road & Connection Point Boulevard

Signal ID	charlotte-5516526369
Location	NPA 52 · 0.85 mi from site
Added PM peak trips	9
Existing / No-Build (PM)	LOS E · 75.2 s/veh · v/c 1.07
Build (PM)	LOS E · 76.9 s/veh · v/c 1.07
Δ control delay	+1.7 s/veh
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	531	0	531	0.66 → 0.66	23.4 → 23.4	C → C	428
SB	444	0	444	0.55 → 0.55	20.7 → 20.7	C → C	334
EB	561	0	561	0.69 → 0.69	24.6 → 24.6	C → C	463
WB	382	9	391	0.47 → 0.48	19.3 → 19.5	B → B	284

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	9	100%

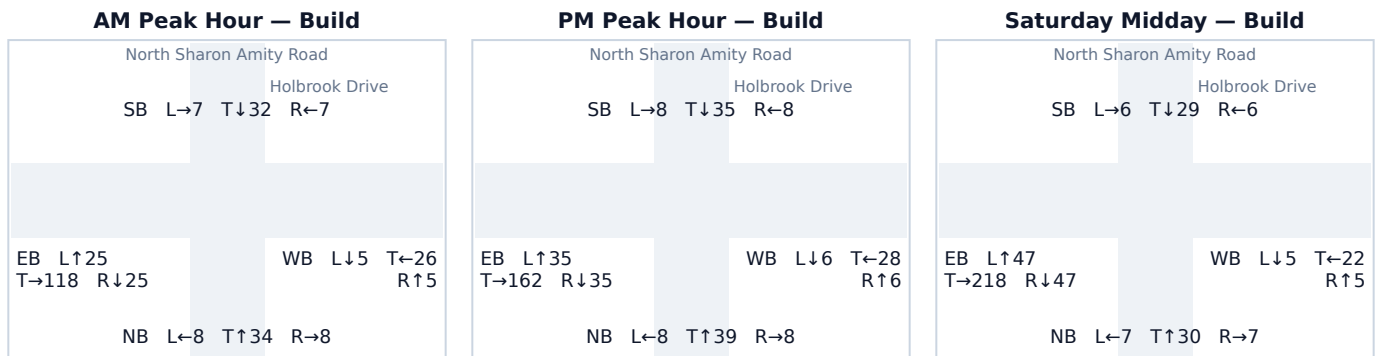
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 North Sharon Amity Road & Holbrook Drive

Signal ID	charlotte-172660814
Location	NPA 16 · 0.02 mi from site
Added PM peak trips	177
Existing / No-Build (PM)	LOS B · 14.6 s/veh · v/c 0.11
Build (PM)	LOS B · 15.6 s/veh · v/c 0.21
Δ control delay	+1.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	55	0	55	0.07 → 0.07	14.2 → 14.2	B → B	32
SB	51	0	51	0.06 → 0.06	14.2 → 14.2	B → B	30
EB	55	177	232	0.07 → 0.29	14.2 → 16.5	B → B	151
WB	40	0	40	0.05 → 0.05	14.0 → 14.0	B → B	23

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	177	100%

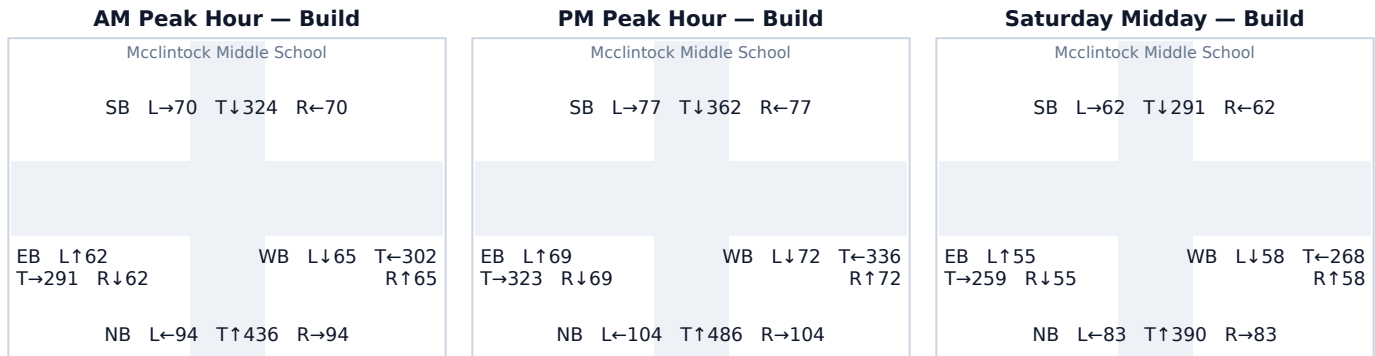
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Mcclintock Middle School

Signal ID	charlotte-cdot-80017
Location	NPA 99 · 0.90 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS F · 123.3 s/veh · v/c 1.19
Build (PM)	LOS F · 124.3 s/veh · v/c 1.20
Δ control delay	+1.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	692	2	694	0.85 → 0.86	33.3 → 33.5	C → C	641
SB	514	2	516	0.63 → 0.64	22.8 → 22.9	C → C	411
EB	461	0	461	0.57 → 0.57	21.2 → 21.2	C → C	352
WB	480	0	480	0.59 → 0.59	21.7 → 21.7	C → C	371

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	2	50%
NB Through	2	50%

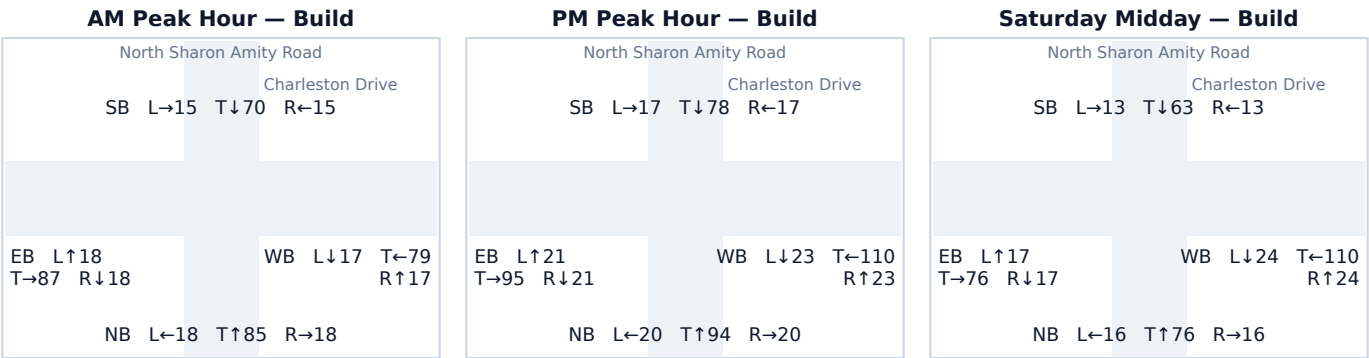
Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 North Sharon Amity Road & Charleston Drive

Signal ID	charlotte-7810838990
Location	NPA 52 · 0.19 mi from site
Added PM peak trips	55
Existing / No-Build (PM)	LOS B · 16.3 s/veh · v/c 0.27
Build (PM)	LOS B · 16.7 s/veh · v/c 0.30
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	134	0	134	0.17 → 0.17	15.2 → 15.2	B → B	82
SB	112	0	112	0.14 → 0.14	14.9 → 14.9	B → B	68
EB	137	0	137	0.17 → 0.17	15.2 → 15.2	B → B	84
WB	101	55	156	0.13 → 0.19	14.7 → 15.4	B → B	97

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	55	100%

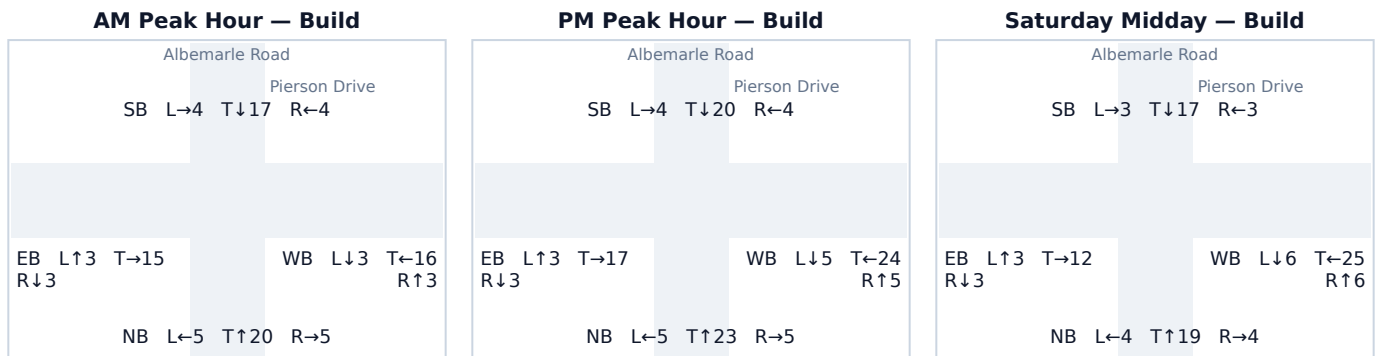
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Albemarle Road & Pierson Drive

Signal ID	charlotte-1711278797
Location	NPA 240 · 0.58 mi from site
Added PM peak trips	17
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	33	0	33	0.04 → 0.04	14.0 → 14.0	B → B	19
SB	28	0	28	0.03 → 0.03	13.9 → 13.9	B → B	16
EB	23	0	23	0.03 → 0.03	13.9 → 13.9	B → B	13
WB	17	17	34	0.02 → 0.04	13.8 → 14.0	B → B	20

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	17	100%

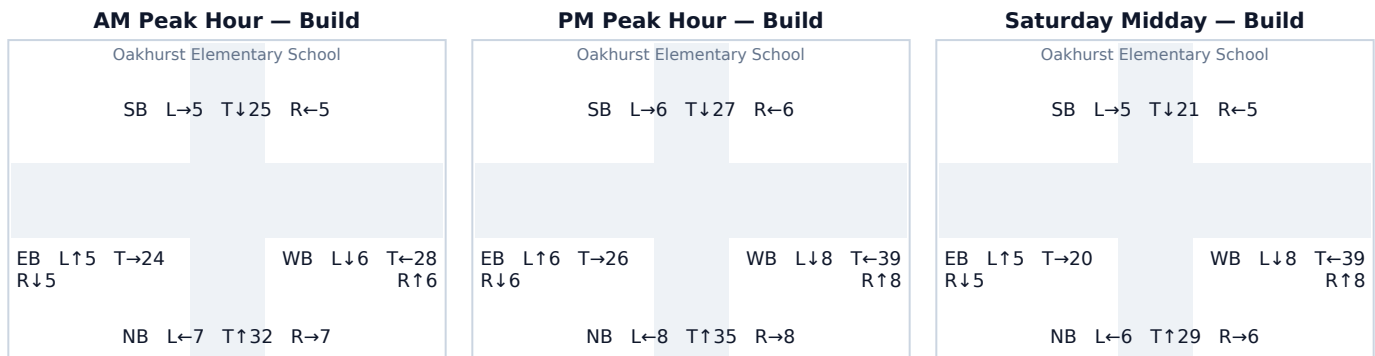
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Oakhurst Elementary School

Signal ID	charlotte-cdot-80097
Location	NPA 240 · 0.97 mi from site
Added PM peak trips	19
Existing / No-Build (PM)	LOS B · 14.4 s/veh · v/c 0.09
Build (PM)	LOS B · 14.5 s/veh · v/c 0.10
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	51	0	51	0.06 → 0.06	14.2 → 14.2	B → B	30
SB	39	0	39	0.05 → 0.05	14.0 → 14.0	B → B	23
EB	38	0	38	0.05 → 0.05	14.0 → 14.0	B → B	22
WB	36	19	55	0.04 → 0.07	14.0 → 14.2	B → B	32

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	19	100%

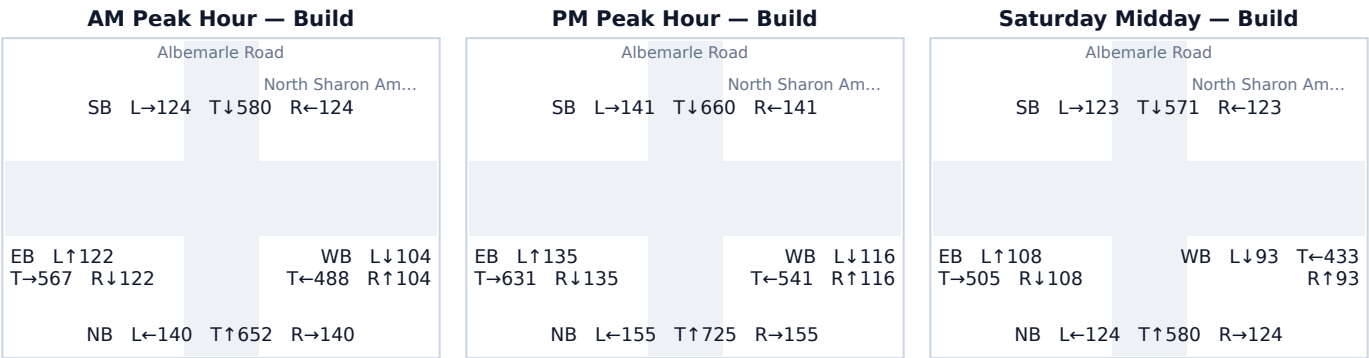
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Albemarle Road & North Sharon Amity Road

Signal ID	charlotte-2181076171
Location	NPA 16 · 0.47 mi from site
Added PM peak trips	89
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.98
Build (PM)	LOS F · 300.0 s/veh · v/c 2.03
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,035	0	1,035	1.28 → 1.28	159.1 → 159.1	F → F	1,059
SB	854	89	942	1.05 → 1.16	71.3 → 111.3	E → F	964
EB	901	0	901	1.11 → 1.11	91.6 → 91.6	F → F	922
WB	773	0	773	0.95 → 0.95	46.2 → 46.2	D → D	769

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	89	100%

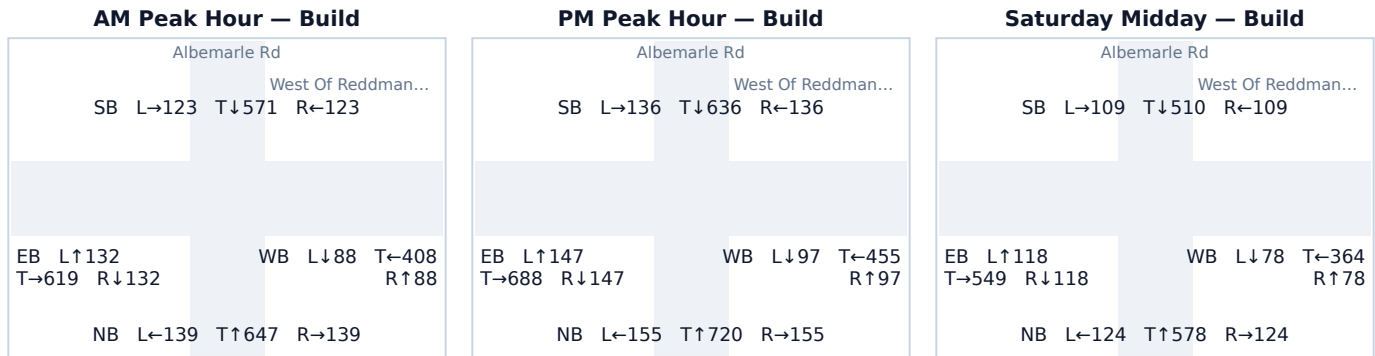
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block

Signal ID	charlotte-cdot-1721
Location	NPA 164 · 0.83 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.98
Build (PM)	LOS F · 300.0 s/veh · v/c 1.98
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,027	3	1,030	1.27 → 1.27	154.6 → 156.4	F → F	1,054
SB	906	2	908	1.12 → 1.12	93.9 → 95.0	F → F	929
EB	982	0	982	1.21 → 1.21	131.2 → 131.2	F → F	1,004
WB	649	1	649	0.80 → 0.80	29.5 → 29.5	C → C	576

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	2	33%
SB Through	2	33%
NB Right	1	17%
WB Left	1	17%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

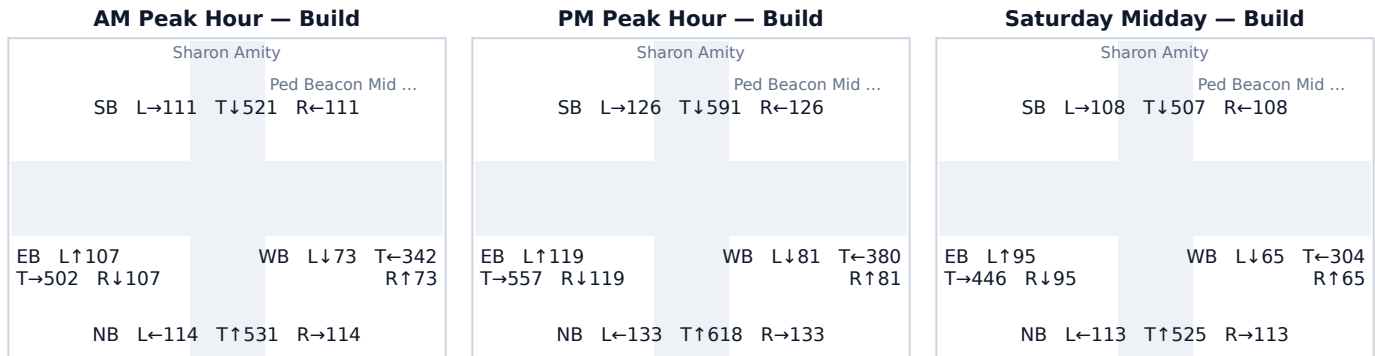
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on

its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)

Signal ID	charlotte-cdot-1614
Location	NPA 164 · 0.86 mi from site
Added PM peak trips	141
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.62
Build (PM)	LOS F · 300.0 s/veh · v/c 1.70
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	811	73	884	1.00 → 1.09	56.3 → 84.0	E → F	904
SB	775	68	843	0.96 → 1.04	46.7 → 67.3	D → E	863
EB	795	0	795	0.98 → 0.98	51.9 → 51.9	D → D	808
WB	542	0	542	0.67 → 0.67	23.9 → 23.9	C → C	440

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	73	52%
SB Through	68	48%

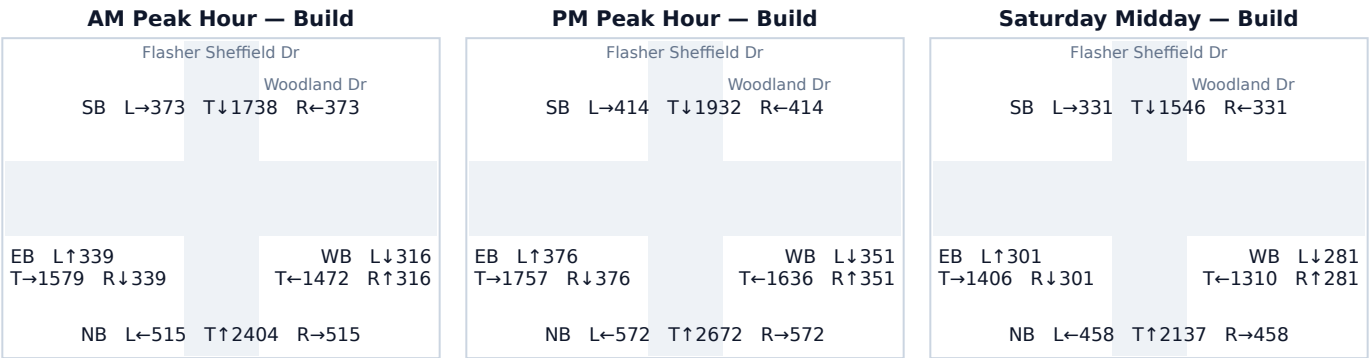
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.14 Flasher Sheffield Dr & Woodland Dr

Signal ID	charlotte-172295413
Location	NPA 365 · 0.93 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 6.34
Build (PM)	LOS F · 300.0 s/veh · v/c 6.35
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	3,816	0	3,816	4.71 → 4.71	300.0 → 300.0	F → F	3,903
SB	2,760	0	2,760	3.41 → 3.41	300.0 → 300.0	F → F	2,823
EB	2,507	1	2,509	3.10 → 3.10	300.0 → 300.0	F → F	2,566
WB	2,337	3	2,338	2.88 → 2.89	300.0 → 300.0	F → F	2,392

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	2	50%
EB Through	1	25%
WB Right	1	25%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Albemarle Rd & Reddman Rd

Signal ID

charlotte-172211825

Location

NPA 164 · 0.98 mi from site

Added PM peak trips

11

Existing / No-Build (PM)

LOS F · 300.0 s/veh · v/c 2.46

Build (PM)

LOS F · 300.0 s/veh · v/c 2.47

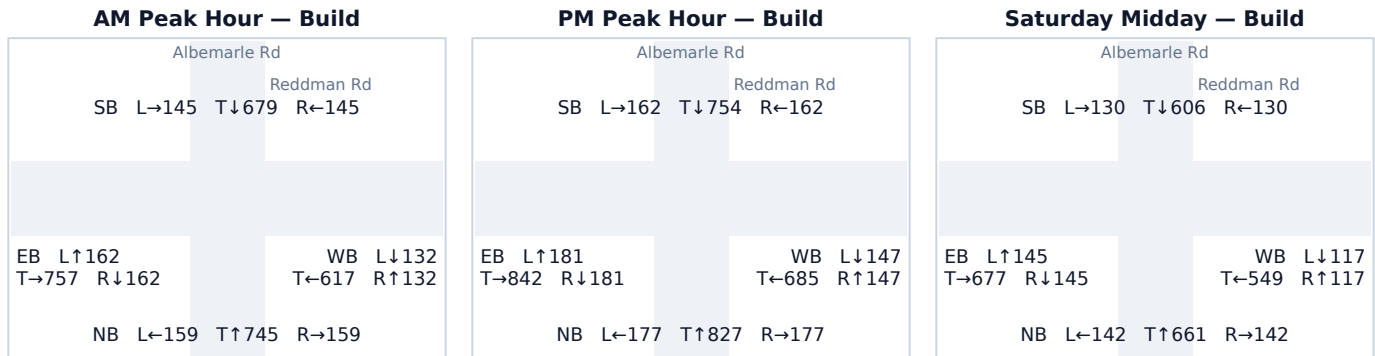
Δ control delay

+0.0 s/veh

Mitigation

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,181	0	1,181	1.46 → 1.46	237.4 → 237.4	F → F	1,208
SB	1,073	5	1,078	1.32 → 1.33	179.0 → 181.9	F → F	1,103
EB	1,198	6	1,204	1.48 → 1.49	246.8 → 250.0	F → F	1,232
WB	979	0	979	1.21 → 1.21	129.9 → 129.9	F → F	1,002

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	6	55%
SB Right	5	45%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.